How to Cut Carbon Fiber on the 3030 Prover Ultra CNC

Safety

Cutting carbon fiber is not safe if you do not control the dust. When carbon fiber is machined, it creates very fine dust particles. These particles can become airborne and can be harmful when inhaled. Because of this, it is very important to prevent the dust from escaping into the workshop.

There are two methods that can be used:

1. Water table method (recommended) - This is the safest option because the water captures most of the dust before it can enter the air.
2. Shop vacuum method - This is easier to set up but less effective. It requires a powerful shop vacuum to continuously remove dust while cutting.

Both methods require a fully enclosed machine to prevent dust and chips from spreading around the room.

Building the Enclosure

A good enclosure is one of the most important parts of safely cutting carbon fiber. The goal is to completely surround the CNC machine and keep dust inside.

Materials

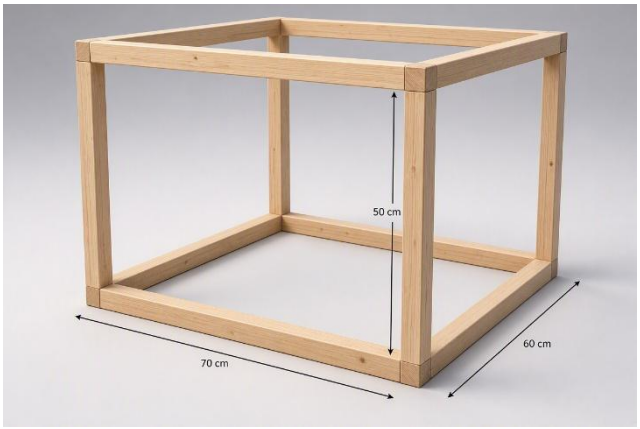
You do not need expensive materials. The enclosure can be built using:

- 28 × 45 mm wooden beams
- Metal L-brackets
- Screws
- Old plastic table covers or thick transparent plastic sheets
- Velcro strips
- Duct tape

If you have a larger budget, clear acrylic sheets can be used instead of plastic covers. Acrylic looks better and is more durable, but it is not required.

Step 1: Build the Frame

Start by building a rectangular frame from the wooden beams. The frame should completely fit around the 3030 Prover Ultra while leaving enough room to work inside.



A minimum recommended size is:

- Length: 70 cm
- Width: 60 cm
- Height: 50 cm

However, making the enclosure slightly larger is strongly recommended. A larger enclosure provides more space for changing tools, clamping material, and cleaning the machine.

Use the L-brackets and screws to connect the beams together. Make sure the frame is sturdy and does not wobble.

Step 2: Cover the Frame



Once the frame is complete, cover all sides with the plastic sheets.

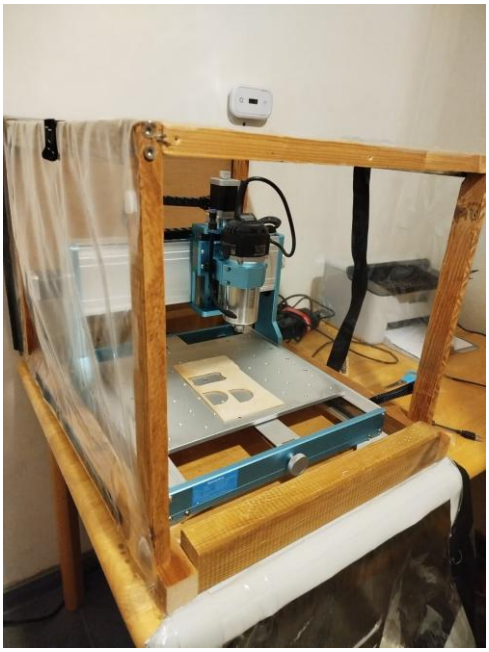
Make sure you cover:

- Front
- Back
- Left side
- Right side
- Top
- Bottom

Many people forget the bottom, but this is important because dust and small chips can fall through openings and end up on the workbench.

Secure the plastic using staples, screws with washers, or strong tape.

Step 3: Add Access Doors



After covering the frame, you still need a way to access the machine.

Choose the front panel and at least one side panel to serve as doors. Carefully cut openings in the plastic and attach Velcro strips along the edges.

This allows you to:

- Open the enclosure when changing tools
- Install or remove material
- Adjust clamps
- Clean the machine

Having only one opening works, but it can become inconvenient.

For the best experience:

- The entire front should open.
- At least one side should open.
- Two opening sides are even better.

This makes it much easier to reach all areas of the machine.

Step 4: Create Cable Passages



The CNC machine requires cables for power, the controller, and sometimes other accessories.

Make a small opening in the back of the enclosure for these cables.

Try to keep the hole as small as possible while still allowing the cables to pass through. Smaller openings reduce the amount of dust that can escape.

After feeding the cables through, seal any gaps with duct tape.

Step 5: Reinforce the Enclosure

To improve durability and dust containment, reinforce the enclosure with duct tape.

Pay special attention to:

- Corners
- Seams between plastic sheets

- Door edges
- Cable openings

Even small gaps can allow fine carbon fiber dust to escape, so take the time to seal everything properly. It's especially important to properly enclose it when using method 2, because in this method there will still be some dust flying around.

Method 1: Water Table (Recommended)

The first method uses a water table in which the carbon fiber sheet is partially submerged while being machined. This is the method used throughout the rest of this guide.

Although this setup requires a little more preparation than simply attaching a vacuum hose, it is by far the safest option. The water captures nearly all of the carbon fiber dust produced during machining, preventing it from becoming airborne and reducing the risk of inhalation. This keeps both the operator and the workshop much cleaner and safer.

One disadvantage of this method is that changing end mills can be slightly more difficult because the work area is surrounded by water. However, with a bit of practice this becomes manageable and is a small trade-off for the significantly improved dust control.

Selecting the Tray

To create the water table, you will need a suitable metal tray.



The tray should meet the following requirements:

- Made from metal, such as galvanized steel or stainless steel.
- Strong enough that it does not bend or deform during use.
- Slightly smaller than the CNC bed.
- At least 30 mm deep, although a deeper tray is preferred.
- Choosing a tray that is slightly smaller than the machine bed makes installation easier and leaves room for the mounting system.

The tray depth is important because it must hold enough water to effectively trap dust while still allowing thicker carbon fiber sheets and fixtures to fit inside. A shallow tray may cause water to splash out during operation and can limit the size of material that can be machined.

An example of a suitable tray is shown above.

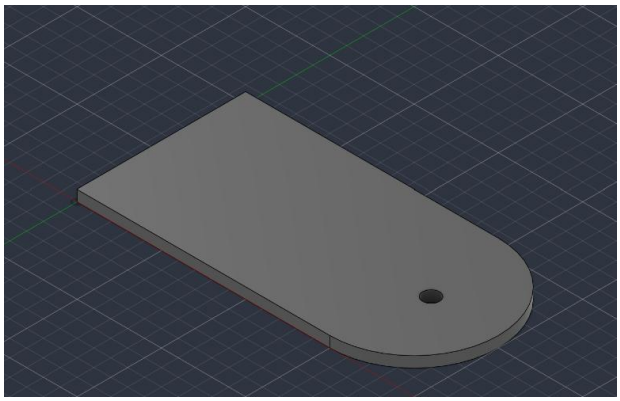
Creating the Mounting System

The water table will be mounted directly onto the CNC machine using the clamping hardware supplied by Genmitsu.

Instead of using standard workpiece clamps, the supplied screws, washers and nuts will be used together with four custom wooden mounting blocks that are attached to the underside of the tray.

These mounting blocks connect the tray securely to the machine while still allowing it to be removed when necessary.

Manufacturing the Mounting Blocks



A CNC design and the required G-code for the mounting blocks are included in the GitHub repository.

1. Search for: "water table mount" to find both the design files and machining files.
2. You will need to machine four identical mounting blocks.
3. Recommended material:
    - Plywood or MDF
    - Approximately 3.2 mm thick

After machining the four mounting blocks, inspect them for burrs or rough edges and make sure they fit correctly with the supplied Genmitsu mounting hardware.

Installing the Mounting Blocks on the CNC Bed

Using the original Genmitsu screws, washers and T-nuts, attach all four mounting blocks to the CNC bed.

At this stage, do not permanently attach the tray yet.

Instead, place the tray on top of the mounting blocks and verify the following:

- The tray sits flat.
- The tray does not contact any screw heads.
- The tray does not interfere with machine movement.
- The tray remains within the machine's working area.

Positioning the Mounts

The location of the four mounting blocks is important.



For maximum stability, the mounts should be positioned as far apart as possible.

Ideally:

- One mount near each corner.
- Even spacing around the tray.
- No mounts grouped together on a single side.

This provides better support and prevents the tray from flexing or shifting during operation.

Take your time during this step because repositioning the mounts later is much more difficult.

Attaching the Tray

1. Once the tray fits correctly and all positions have been verified, remove the tray from the machine.
2. Apply CA glue (cyanoacrylate glue) to the top of each mounting block.
3. Carefully lower the tray back into position and press it firmly onto the four blocks.
4. Make sure the tray is aligned correctly before the glue cures.
5. If you want the glue to set more quickly, you can spray a CA glue accelerator onto the joints. This usually hardens the adhesive almost instantly.
6. Hold the tray in position for a few moments to ensure a strong bond.

Final Result

After the glue has cured, the water table is complete.

Method 2: Shop Vacuum with Dust Shoe

A second option for machining carbon fiber is to use a shop vacuum together with a dust shoe. This method is simpler to set up than a water table and allows easier access to the workpiece and cutting tools. However, it is important to understand that it is less effective at controlling dust.

A dust shoe captures the majority of the dust and chips generated during machining and directs them into the shop vacuum. While this significantly reduces airborne dust, it does not completely eliminate it. Small dust particles may still escape into the surrounding environment.

Because of this, it is strongly recommended to:

- Wear safety glasses or other eye protection.

- Wear a properly fitted respirator suitable for fine dust.
- Perform the machining operation outdoors whenever possible.
- Use an enclosure around the machine to help contain any remaining dust.

Although this method is less safe than using a water table, it can still be a practical solution when a water table is not available.

Choosing a Dust Shoe

A dust shoe is an attachment that mounts around the spindle and surrounds the cutting tool with a brush skirt. As the vacuum draws air through the dust shoe, dust and chips are collected as close to the cutter as possible.

For the Genmitsu 3030 Prover Ultra, SainSmart offers a compatible dust shoe that can be purchased directly from their website:

https://www.sainsmart.com/products/genmitsu-cnc-dust-shoe-for-proverxl-4030

Before ordering, verify that the dust shoe is compatible with your spindle configuration and machine setup.

Installing the Dust Shoe

Installation is generally straightforward and follows the manufacturer's instructions.

The dust shoe is mounted around the spindle and connected to a shop vacuum using a flexible vacuum hose. Once installed, the brush skirt should lightly contact the surface of the workpiece while still allowing the spindle to move freely.

Before beginning any machining operation, manually move the machine through its full range of travel to verify that:

- The dust shoe does not interfere with machine movement.
- The vacuum hose does not become caught on the frame.
- The hose has enough slack to follow the machine during operation.
- The spindle can reach all areas of the workpiece.

Taking a few minutes to check this beforehand can prevent interrupted jobs or accidental damage.

Modifying the Enclosure

If you are using the enclosure described earlier in this guide, one additional modification will be required.

A hole must be made in one side or at the rear of the enclosure to allow the vacuum hose to pass through.

When creating this opening:

- Keep the hole only slightly larger than the hose diameter.
- Position it where the hose can move freely.
- Ensure it does not interfere with machine movement.

After routing the hose through the enclosure, seal any large gaps around it using duct tape or flexible foam. This helps prevent dust from escaping through the opening.

Connecting the Vacuum Hose

In many cases, the diameter of the shop vacuum hose will not match the connection on the dust shoe.

To solve this problem, you may need to use:

- A commercially available hose adapter.
- A custom CNC-machined adapter.
- A 3D-printed adapter.
- A 3D-printed adapter is often the easiest solution, as it can be designed to perfectly fit both the dust shoe and the vacuum hose.

The adapter should create a secure connection while allowing sufficient airflow for effective dust extraction.

Operating the System

Once everything is assembled:

1. Secure the carbon fiber sheet to the machine.
2. Close the enclosure.
3. Turn on the shop vacuum.
4. Verify that airflow is present at the dust shoe.
5. Start the CNC program.
6. Leave the vacuum running until all dust has been collected.

After machining is completed, allow the vacuum to run for a short period before opening the enclosure. This gives any remaining airborne particles time to be extracted.

Bit selection

Choosing the right cutting tool is one of the most important factors when machining carbon fiber. While it is technically possible to use the same end mills that are normally used for wood, plastics, or aluminum, this is generally not recommended.

Carbon fiber is an extremely abrasive material. The carbon fibers themselves act almost like tiny cutting tools and continuously wear away the cutting edges of the end mill. A tool that can last for many hours when cutting wood may become noticeably dull after only a relatively short amount of time when machining carbon fiber.

Because of this, corn-style end mills are usually recommended for hobby-level carbon fiber machining. The many small cutting edges distribute the cutting forces more evenly, help break up chips, and tend to tolerate the abrasive nature of carbon fiber better than standard end mills. They are also relatively inexpensive, making them a practical choice for most users.

For occasional hobby use, standard carbide corn bits are generally more than sufficient. However, if you plan to machine large quantities of carbon fiber or produce parts regularly, it may be worth investing in specialized composite-cutting tools. These tools are specifically designed for materials such as carbon fiber, fiberglass, and other composite laminates. They typically provide longer tool life, cleaner edge quality, and reduced delamination.

Regardless of which tool you choose, it is important to inspect it regularly for wear. A dull tool will generate more heat, produce poorer cut quality, increase cutting forces, and place additional load on the machine.

Durability of Corn Bits

One advantage of corn bits is their surprising durability.

In my own experience, I have accidentally driven two corn bits completely through a 5 mm carbon fiber sheet, through the water table, and into the aluminum machine bed. Despite this rather severe crash, both tools survived without breaking.

This does not mean that crashes are harmless, however.

If a crash occurs while using a water table:

1. Immediately stop the machine.
2. Turn off all electrical power.
3. Remove water from the work area as quickly as possible.
4. Inspect the machine before restarting.

Carbon fiber is electrically conductive, and when mixed with water it creates a conductive slurry that can easily reach electronics, motors, connectors, and other sensitive machine components. Fast action can prevent expensive damage.

Recommended Tool Sizes

For most carbon fiber projects, especially drone frames, I primarily use 1 mm to 2 mm diameter corn bits.

These sizes offer a good balance between strength and detail:

- Small enough to machine tight corners and intricate features.
- Strong enough to tolerate reasonable cutting loads.

- Capable of achieving approximately 1 to 2 mm depth of cut in many situations.

A larger cutter will generally allow higher material removal rates but may struggle with small internal features. A smaller cutter can produce finer details but is more fragile and often requires more conservative cutting parameters.

For most hobby projects, a 1 mm or 2 mm corn bit is an excellent starting point.

Chamfering and Countersinking

After machining, it is often desirable to chamfer holes or edges.

For this purpose, a normal chamfering bit can be used without any special modifications. However, because carbon fiber is highly abrasive, chamfer bits should also be checked frequently for wear.

Whenever possible, only use the chamfer tool when necessary. Excessive chamfering can quickly dull the cutting edges and increase tooling costs.

A worn chamfer tool can leave rough edges, poor surface finishes, and inconsistent chamfers.

Upcut Tools and Water Splashing

All of my testing has been performed using **upcut-style cutters**.

For corn bits, this generally does not create any significant problems.

Chamfer bits, however, can behave differently. Their geometry often has a tendency to throw water away from the cutting area while spinning. This can result in:

- Increased splashing.
- More water inside the enclosure.
- Reduced visibility of the cutting process.
- Additional cleanup after machining.

To reduce this problem, I recommend designing and 3D printing a simple protective sleeve or cylinder that surrounds the chamfer bit while still leaving enough clearance for machining operations.

This shield helps contain water close to the cutting area and prevents excessive splashing throughout the enclosure.

The design does not need to be complicated. Even a simple cylindrical guard can significantly improve the user experience when chamfering parts in a water table setup.

Generating Toolpaths

Generating toolpaths for carbon fiber is very similar to generating toolpaths for any other material. You can use your preferred CAM software and workflow. Personally, I use Fusion 360, but any CAM package that allows proper control over feeds, speeds, depths of cut, and tool selection will work.

To make getting started easier, I have included my personal tool library with feeds and speeds in the GitHub repository. These values should be treated as a starting point rather than absolute rules. Every machine setup is slightly different, and factors such as spindle power, workholding, tool wear, and material quality can all affect the optimal settings.

A good approach is to start conservatively and gradually increase the feeds and speeds while paying close attention to the machine. Listen carefully during machining. If you begin to hear excessive vibration, chatter, or signs that the spindle is struggling, reduce the cutting parameters. Cutting carbon fiber should produce a consistent and controlled cutting sound rather than loud vibrations or aggressive chatter.

Tool Changes When Using a Water Table

One complication of using a water table is tool changes.

Under normal circumstances, after changing a tool you would simply touch off or probe the top of the workpiece again. However, when the carbon fiber sheet is submerged or partially submerged in water, probing directly on the material becomes inconvenient and can be unreliable.

A much better solution is to create a fixed-height reference block on the machine that will be used for all future tool changes.

Step 1: Zero the First Tool

Install the first tool that will be used in the job.

With the water table installed and the carbon fiber securely mounted, probe or manually set the Z-zero on the top surface of the carbon fiber as you normally would.

This establishes the correct work coordinate system for the entire job.

After setting Z-zero, do not change the work coordinate system again.

Step 2: Create a Fixed Reference Block

Next, securely mount a block of known height somewhere on the machine bed where it will not interfere with machining.

This block can be:

- Aluminum
- Plastic
- Hardwood
- Any stable material

The exact material is not important, but it must remain fixed throughout the entire machining process.

- Choose a location that:
- Is easy to access.
- Is outside the machining area.
- Will not collide with clamps or fixtures.
- Remains dry if possible.

Once mounted, the block should never move until the entire job has been completed.

Step 3: Record the Reference Height

Before beginning the program, move the first tool above the reference block. Slowly lower the tool until it just touches the block. Record the machine's Z-position value.

For example:

Tool 1 touches the reference block at Z = -35.00 mm

Write this number down because it will serve as your reference measurement.

Step 4: Change the Tool

When a tool change is required:

- Pause the program.
- Raise the spindle.
- Replace the tool.

Move above the reference block.

Then slowly lower the new tool until it just touches the same reference block.

Suppose the new tool touches at:

Tool 2 touches the block at Z = -37.20 mm

Step 5: Calculate the Difference

Now compare the new measurement to the original measurement.

Example:

Original tool: -35.00 mm

New tool: -37.20 mm

Difference:

2.20 mm

This means the new tool is sticking out 2.20 mm further than the original tool.

Step 6: Apply the Offset

There are several ways to do this depending on the controller you are using.

The general idea is simple:

- If the new tool sticks out further, raise the Z-zero by the measured difference.
- If the new tool is shorter, lower the Z-zero by the measured difference.

Most controller software allows you to directly adjust the work offset or work coordinate system value.

After applying the offset, the machine will know where the top of the carbon fiber sheet is again without requiring you to probe the submerged workpiece.

Always perform a careful test move after applying the offset to verify everything is correct before restarting the spindle.

Keep Toolpaths Reasonably Short

When using the water table system described in this guide, the tray is attached using glued mounting blocks.

Although this setup is surprisingly strong, water, vibrations, and long machining times can eventually weaken the adhesive bond.

For this reason, it is generally a good idea to split large jobs into smaller operations whenever possible.

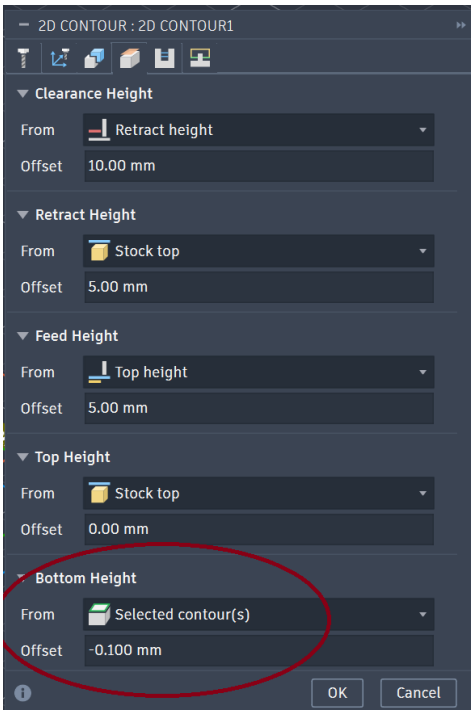
Instead of running several hours of uninterrupted machining, consider:

- Cutting profiles first.
- Checking the setup.
- Verifying that everything remains secure.
- Continuing with the next operation.

Occasionally inspect the water table during machining to ensure that nothing has moved or loosened.

Use Sufficient Step-Down Below the Material

When cutting completely through carbon fiber, it is important to program enough additional depth below the material thickness.



Many people are tempted to cut exactly to the measured thickness of the material, but this often causes problems.

For example:

- Measured material thickness: 5.0 mm
- Programmed cut depth: 5.0 mm

If the material has even a slight tilt or unevenness, some areas may not cut completely through.

Instead, add some additional depth.

For example:

- Material thickness: 5.0 mm
- Final cut depth: 5.3 mm to 5.5 mm

This ensures the cutter completely separates the part from the stock even if the material is not perfectly flat.

Measure Every Carbon Fiber Sheet

One mistake many beginners make is assuming that carbon fiber sheets are exactly the thickness listed by the manufacturer.



In reality, they often vary slightly.

A sheet sold as: 2 mm might actually measure:

- 1.9 mm
- 2.05mm
- 2.1mm
- Or if you got lucky 2.00mm

depending on manufacturing tolerances.

Because of this, you should always measure the actual material thickness with calipers before creating your CAM setup.

This provides several advantages:

- More accurate pocket depths.
- Better-fitting parts.
- Cleaner through-cuts.
- More reliable chamfers and countersinks.

Taking a quick measurement only takes a few seconds and can prevent many machining problems later in the process.

Workholding

Proper workholding is extremely important when machining carbon fiber. Even with perfect feeds and speeds, a loose workpiece can ruin a part, damage a tool, or in the worst case cause a dangerous situation if a piece of carbon fiber comes loose during machining.

The workholding method depends on which dust-control method you are using.

Method 2: Dust Shoe and Shop Vacuum

If you are using Method 2, where a dust shoe and shop vacuum are used, workholding is exactly the same as when machining plywood or other sheet materials.

You can use:

Painter's tape and CA glue.

Double-sided tape.

Clamps.

Vacuum fixtures.

Any other workholding method that you normally use on the CNC.

Because there is no water involved, the standard painter's tape and CA glue method usually works very well and does not require any special modifications.

Method 1: Water Table

When using Method 1, things become slightly more complicated.

The commonly used painter's tape and CA glue method still works, but water can weaken the bond over time. During longer machining operations, the tape may begin to lift, and this can allow parts to move or even come loose completely.

To make the setup more reliable, I developed a modified version of the painter's tape and CA glue method that provides a much stronger hold while still being relatively easy to remove afterwards.

Step 1: Clean the Water Table

Before applying any tape or glue, thoroughly clean the surface of the metal water table.

I recommend using isopropyl alcohol and a clean cloth.

This removes:

- Oil.
- Dust.
- Dirt.
- Manufacturing residue.

A clean surface is essential for achieving a strong bond.

Allow the surface to dry completely before continuing.

Step 2: Attach Painter's Tape to the Water Table

Apply strips of painter's tape to the top surface of the metal water table.

Try to cover as much of the contact area as possible.

Ensure that:

- The tape is pressed down firmly.
- There are no air bubbles.
- The tape lies flat against the metal.

The better the tape adheres, the stronger the final connection will be.

Step 3: Attach the Spoilboard

Apply CA glue to the painter's tape on the water table.

Next, spray CA glue accelerator onto the underside of the spoilboard.

Carefully position the spoilboard above the water table and press it firmly into place.

Hold pressure for several seconds.

The accelerator will cause the glue to cure almost immediately, creating a strong bond between the spoilboard and the water table.

At this point, the spoilboard should feel completely rigid and should not move when moderate force is applied.

Step 4: Prepare the Carbon Fiber Sheet

Before attaching the carbon fiber, clean its surface with isopropyl alcohol.

Carbon fiber sheets often contain:

- Dust.
- Fingerprints.
- Mold-release residue from manufacturing.

These contaminants can reduce adhesive strength.

Allow the surface to dry completely before proceeding.

Step 5: Apply Painter's Tape to the Carbon Fiber

Apply painter's tape to the underside of the carbon fiber sheet.

Press the tape down firmly to ensure maximum adhesion.

Just like before, try to eliminate any air bubbles or loose edges.

The tape should fully contact the carbon fiber surface.

Step 6: Attach the Carbon Fiber to the Spoilboard

Apply CA glue onto the painter's tape attached to the carbon fiber sheet.

Next, spray accelerator onto the surface of the spoilboard.

Carefully position the carbon fiber sheet and lower it onto the spoilboard.

Once properly aligned, apply firm and even pressure for several seconds.

The glue should cure quickly and produce a surprisingly strong bond.

The result is a setup that is much more resistant to water exposure than a standard painter's tape and glue setup. As it only has 2 layers of painters tape instead of 4.

Machining

With the enclosure built, the water table installed, the workpiece secured, and the toolpaths generated, you are finally ready to start machining carbon fiber.

Although the actual cutting process is relatively straightforward, there are a few important precautions that should be taken before pressing the start button. Taking a few extra minutes to check everything can save you from broken tools, damaged fixtures, and potentially expensive repairs.

Preparing the Machine

Before filling the water table, start your CNC control software and connect it to the machine.

At this stage, the water table should still be empty.

One of the first things you should do is set your work coordinates and probe the surface of the material. It is much easier and safer to perform probing operations before water is added to the setup.

Once the work coordinates have been established and verified, you can fill the water table with water.

Doing it in this order helps prevent accidental spills and makes it easier to access the workpiece during setup.

Remove the Controller Screen

Before adding water, it is also recommended to remove the small controller screen from the machine.

While water splashing is normally limited, accidents can happen. During machining, tool changes, or setup mistakes, water may be thrown around the enclosure. The last thing you want is water reaching any electronic components.

Removing the screen only takes a few seconds and eliminates one potential point of failure.

It is better to move it to a safe and dry location before beginning the job.

Always Check the G-Code Preview

Before starting any machining operation, load your G-code into the preview window of your CNC controller software.

Personally, I use UGS (Universal Gcode Sender), but the same principle applies regardless of which sender you use.

Never assume that the generated G-code is correct.

Always inspect:

- Travel moves
- Retract heights
- Tool changes
- Drilling operations
- Lead-ins and lead-outs
- Final cutting depths

Pay particular attention to any movement in the Z-axis.

A few minutes spent checking the simulation can prevent hours of repairs.

My Experiences with Unexpected Crashes

During the development of this project, I managed to crash tools into the water table twice.

Fortunately, neither crash caused permanent damage, but both incidents could have been avoided.

In both cases, the cause was not the CAM software, but rather something unexpected happening after the G-code had already been generated.

Problem 1: Decimal Separator Issues

The first crash happened because of European number formatting.

In many European countries, decimal values are written with a comma.

For example:

23,00

However, some CNC software expects periods instead of commas.

For example:

23.00

When entering an offset value, I accidentally used:

23,00

UGS interpreted this incorrectly and effectively treated the value as a much larger number. As a result, my offset became completely wrong and the machine attempted to move directly into the bed.

The lesson here is simple:

Always verify any manually entered coordinate offsets before running the machine.

A single typing mistake can completely change the tool position.

Problem 2: Auto-Leveling Changes the G-Code

The second issue occurred while using the UGS Auto Leveler.

The purpose of auto-leveling is to compensate for slight variations in surface height by modifying the G-code based on measurements taken across the workpiece.

In theory, this is very useful, especially when precise Z-height control is required.

However, on two separate occasions I noticed that the generated toolpath had been modified incorrectly.

After processing, the toolpath contained plunge moves that would drive the cutter directly into the spoilboard or machine bed.

Fortunately, these unexpected moves were visible in the G-code preview.

Because I checked the preview before machining, I caught the problem before starting the spindle.

Both occurrences looked very similar and happened during testing, which makes me suspect there may be a bug or edge case within the software.

Be Careful When Using Auto-Leveling

I am not saying that the UGS Auto Leveler should never be used.

In fact, it can be extremely useful when:

- High dimensional accuracy is required.
- The stock surface is slightly uneven.
- Precise engraving operations are being performed.

However, whenever using auto-leveling, I strongly recommend visually checking the processed G-code before running it.

Specifically look for:

- Unexpected vertical plunges.
- Extremely deep cuts.
- Rapid moves below the material surface.
- Toolpaths that suddenly dive into the spoilboard.

If anything looks suspicious, investigate it before starting the machine.

Never trust modified G-code blindly.

My Preferred Approach

Personally, I usually avoid auto-leveling when cutting carbon fiber.

Instead, I simply cut slightly deeper into the spoilboard.

For example, rather than trying to stop exactly at the bottom of a 5 mm sheet, I intentionally allow the cutter to continue a small distance into the spoilboard.

This approach:

- Simplifies setup.
- Eliminates the need for auto-leveling.

- Guarantees complete cut-through.
- Reduces the chance of uncut sections remaining attached.

Although it sacrifices a small amount of spoilboard material, spoilboards are inexpensive and easy to replace. And you loose quite a bit of z accuracy.

For my workflow, this trade-off is well worth it.

During Machining

Once the machine is running:

- Keep the enclosure closed.
- Monitor the machining process.
- Listen for unusual sounds.
- Check that the workpiece remains firmly attached.
- Watch for excessive vibration or chatter.
- Verify that water remains inside the tray.
- Be prepared to stop the machine immediately if something looks wrong.

Remember that carbon fiber machining generally produces very little visible debris when using a water table. Most of the dust is trapped by the water, which is exactly what we want. But this also means that you will not be able to see the workpiece, always stay with the machine use eye protection and hearing protection. Have your hand ready on the ESTOP even if you already ran this programming! These machines are very fun to use but can cause some damage if anything goes wrong.

Final Thoughts

Carbon fiber machining on the 3030 Prover Ultra is surprisingly straightforward once the setup is complete. Most mistakes do not happen during cutting itself, but during setup, probing, offsetting, or G-code preparation.

For that reason, the most important habit you can develop is simple:

Always inspect the toolpath preview before pressing start.

A quick visual check can reveal incorrect offsets, unexpected plunges, software errors, and many other potential problems before they turn into broken tools or damaged equipment.

A final word of advice when you are postprocessing the carbon fiber parts, do this outside and wear a resperator or do this inside a bucket of water. Your loungs aren't worth sacrificing. For the design files of the frame and the GCode look in the Github repo.